

Haizhou Yu

✉ szz5ft@virginia.edu | ☎ 4344220970 | 📍 17343 W Copper Lakes Dr, Houston TX | 🌐 <https://www.linkedin.com/in/haizhou-yu-5416b7241/>

Experience

PACCAR

Lewisville, Texas, United States
May 2024 - Aug 2024

Diagnostic System Engineer Intern

- Automated diagnostic report downloads using Python Selenium, cutting manual collection effort by ~80%.
- Helped build a sprint analysis dashboard using PowerBI and DevOps data, improving team efficiency and feedback.
- Developed SQL + PowerBI pipeline for analyzing CAN DTC data for 10000+ vehicles, improving diagnostic reporting and failure analysis.
- Programmed manufacturing sequence for new vehicle architecture. Transferred system to new data model, **improving efficiency by 50%**.
- Gained experience in agile SDLC, actively participating in Scrum, Triage, and other team meetings.
- Implemented features and resolved defects in **DAVIE4 diagnostic software**, improving reliability with next-gen **vehicle architecture**.

Cavalier Autonomous Racing

Charlottesville, Virginia, United States
May 2025 - May 2026

Vehicle Dynamics and Control Engineer, Motion Planning Engineer

- Developed new vehicle tire dynamics models (tire slip, traction limits) from telemetries recorded from 10+ laps to improve simulation fidelity.
- Responsible for simulation development. Implemented mcap overlay player to overlay data to simulation data.
- Developed Python tools for GGV diagram generation and raceline speed optimization to support autonomous racing simulation.
- Participated in testing at WeatherTech Raceway Laguna Seca for 14 days, collecting telemetry and validating vehicle control models.

Streamax America LLC

Houston, Texas
July 2026 - Present

Technical Support Engineer

- Performed on-site quality control, functional validation of recorders, identifying abnormal product and escalating technical issues.
- Troubleshoot product hardware, software, and system-level behavior while developing working knowledge of MDVR architecture, vehicle integration, and CAN-based functionality.
- Developed a **Chrome extension for dashcam onboarding**, providing contextual guidance and configuration instructions within the device's local web interface to improve technician and customer usability.

Education

University of Virginia; Bachelor of Science, Computer Engineering

August 2021 - May 2025

GPA 3.74. Awarded "Dean's List". Major Courses Taken: Data Structures&Algorithms, Computer Systems, Software Development, Digital Logic Design, Embedded Programming, Cybersecurity, Computer Architecture, Autonomous Vehicles. Teaching Assistant for Computer architecture

University of Virginia; Master of Engineering, Computer Engineering

August 2025 - May 2026

GPA 3.85, Courses taken: AI Hardware, Advance Embedded Systems, Business & Technology Leadership of Engineers, Wireless IoT. Teaching assistant for Embedded Systems Design

Projects

BLE Lost and Found Tagger - NRF52840

Jan 2024 - May 2024

- Designed a low-power **BLE tracking device** with **NFC tap-to-connect** functionality for recovering lost personal items
- Developed **embedded firmware for BLE communication and NFC proximity alerts** sharing contact information.

Smart MIDI Controller – Embedded System Capstone - Raspberry Pico, 3D Printing

Dec 2024

- Led a 5-person engineering team to design and build a **custom MIDI hardware controller** for beginner music production
- Designed **PCB and embedded firmware** enabling real-time MIDI signal processing and control with 17 inputs
- Modeled enclosure and mechanical components in **3D CAD**, prototyped using **3D printing**
- Integrated hardware, firmware, and GUI interface into a **complete interactive music system with latency less than 50ms**
- Conducted component sourcing, circuit design, and hardware debugging during prototype iteration

F1TENTH Autonomous Racing Platform

Dec 2024

- Developed autonomous control stack for 1/10-scale racecar using **ROS and LiDAR sensing**
- Implemented **PID-based steering and velocity control** for stable high-speed driving of up to 20km/h
- Constructed **optimal raceline trajectory** from track data to maximize cornering speed
- Programmed 360 degree, 10Hz **LiDAR-based obstacle detection and avoidance algorithms**
- Integrated localization, planning, and control modules for real-time autonomous racing

Real-Time Embedded System | TM4C123G, C, RTOS

Jan 2026 - May 2026

- Built a multithreaded embedded application on the **ARM Cortex-M4–based TM4C123G**, integrating real-time joystick input, LCD rendering, collision detection, and asynchronous game-state updates.
- Designed concurrent RTOS tasks for dynamically generated moving objects and used **semaphores/critical sections** to coordinate shared display resources and prevent race conditions.
- Implemented **PWM peripheral control**, pseudorandom behavior, thread lifecycle management, and deadlock prevention for real-time tasks.

Skills

Programming and Embedded Systems

SPI, I2C, PWM, UART, STM32, J1939, CAN, C, C++, Python, MATLAB, ROS, Java, Assembly, Verilog, BLE, NFC, LoRA, 802.11, HTML, JSON, NumPy, YAML, XML, Pandas, Linux, CMake, Network Architecture

Engineering Skills

PowerBI, Database, SQL, Excel, Matlab, EDA, Multisim, Ultiboard, Data analytics, CANalyzer, dSPACE, ADAS, Fusion 360, Unity, CAD, Game Development, Machine Learning